

Analysis of Mortality on Anonymized, Simulated Data

Simulated within sample

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Experimental Design and Analysis

Anopheles quadrimaculatus third instar larvae were exposed to **TREATMENT** larvicide. Mortality was assessed at 24, 48, 72, and 96 hours, with 4 replicates each ($n = 25$ larvae per replicate). This report contains assay results for a simulated dataset.

Analytic Methods

All analyses were completed in R ver. 4.5.2¹. All comparisons were performed with a Fisher's Exact Test, as expected cell means were < 5 ². For all statistical tests, significance was assessed at $\alpha = 0.05$ (i.e., $p\text{-value} < \text{or} = 0.05$ demonstrated a significant difference).

Mortality of control replicates

Mortality within control replicates was calculated. Mortality of controls should be $< 10\%$. Replicates with mortality = or $> 10\%$ were identified as outliers.

¹R Core Team (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.

²Zar, J. H. 2009. *Biostatistical Analysis* (5th ed.). Prentice Hall: Upper Saddle River, NJ, USA.

Analysis of treatment replicates

A two-sided Fisher's Exact Test was used to perform pairwise comparisons of mortality for treatment replicates. Replicates that were significantly different from others were identified as outliers. The Holm method was used to adjust p-values for multiple comparisons. In situations where two of the replicates were significantly different from the remaining two replicates, the replicates with higher mortality values were treated as outliers.

Comparison of treatment to control

A one-sided Fisher's Exact Test was used to determine if mortality for the treatment group was greater than the control. When outliers were detected in either control or treatment replicates, the test was repeated excluding the outliers.

Synopsis

Mortality of control replicates

Mortality of control replicates was $< 10\%$ for all durations considered in week 1. Some replicates from weeks 2, 3, 4, 5, and 6 had mortality $>$ or $= 10\%$.

Analysis of treatment replicates

There were no significant differences among treatment replicates for weeks 2, 5, and 6. There were significant differences among replicates for week 1 (replicate 1 at 48 hours treated as an outlier), week 3 (replicate 4 at 48 hours and replicate 1 at 96 hours treated as outliers), and week 4 (replicate 2 at 48 hours treated as an outlier).

Comparison of treatment to control

For all weeks, mortality in the treatment group was significantly greater than mortality in the control group, regardless of the presence of outliers in the dataset.

Week 1 results

For week 1, there were four replicates (1, 2, 3, 4) for both the control and treatment groups. Mortality was assessed at 24 and 48 hours.

Mortality of control replicates

For all control replicates in week 1, mortality was $< 10\%$.

Analysis of treatment replicates

During week 1, readings taken at 48 hours had significantly different mortality between replicates.

Readings at 48 hours: Replicate 1 (68% mortality) was significantly different from replicates 2 (100% mortality) and 4 (100% mortality). Replicate 1 was treated as an outlier.

Duration	Replicates for comparison	Adjusted p-value
48hr	1, 2	0.0242
48hr	1, 4	0.0242

Comparison of treatment to control

Mortality was significantly greater in the treatment than in the control for each experiment duration (24 and 48 hours) in week 1.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	4	92	< 0.001
48hr	7	94	< 0.001

Comparison of treatment to control (excluding outliers)

Treatment replicate 1 from 48 hours was treated as an outlier and removed. Mortality was significantly greater in the treatment than in the control for each experiment duration (24 and 48 hours) in week 1.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	1.3	92	< 0.001

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
48hr	0.0	92	< 0.001

Week 2 results

For week 2, there were four replicates (1, 2, 3, 4) for both the control and treatment groups. Mortality was assessed at 24, 48 and 72 hours.

Mortality of control replicates

During week 2, the following control replicates had mortality $\geq 10\%$:

Readings at 24 hours: Control replicate 3 had 28% mortality. This replicate was treated as an outlier.

Duration	Replicate	Dead	Total	%Mortality
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Analysis of treatment replicates

There were no treatment replicates for week 2 that were significantly different from other treatment replicates.

Comparison of treatment to control

Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 48, and 72 hours) in week 2.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	0	94	< 0.001
48hr	2	97	< 0.001
72hr	2	83	< 0.001

Comparison of treatment to control (excluding outliers)

Control replicate 3 from 24 hours was treated as an outlier and removed. Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 48, and 72 hours) in week 2.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	0	94	< 0.001
48hr	2	97	< 0.001

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
72hr	2	83	< 0.001

Week 3 results

For week 3, there were four replicates (1, 2, 3, 4) for both the control and treatment groups. Mortality was assessed at 24, 48, 72, 96 hours.

Mortality of control replicates

During week 3, the following control replicates had mortality $\geq 10\%$:

Readings at 48 hours: Control replicate 2 had 16% mortality. This replicate was treated as an outlier.

Duration	Replicate	Dead	Total	%Mortality
72hr	1	3	25	12
72hr	4	3	25	12

Analysis of treatment replicates

During week 3, readings taken at 48 and 96 hours had significantly different mortality between replicates.

Readings at 48 hours: Replicate 4 (64% mortality) was significantly different from replicate 1 (100% mortality) and replicate 2 (100% mortality). Replicate 4 was treated as an outlier.

Readings at 96 hours: Replicate 1 (64% mortality) was significantly different from replicate 2 (100% mortality), replicate 3 (96% mortality) and replicate 4 (100% mortality). Replicate 1 was treated as an outlier.

Duration	Replicates for comparison	Adjusted p-value
48hr	1, 4	0.00978
48hr	2, 4	0.00978
96hr	1, 2	0.00978
96hr	1, 3	0.04240
96hr	1, 4	0.00978

Comparison of treatment to control

Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 28, 72, and 96 hours) in week 3.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	1	87	< 0.001
48hr	0	99	< 0.001
72hr	7	91	< 0.001
96hr	1	82	< 0.001

Comparison of treatment to control (excluding outliers)

Control replicate 2 from 48 hours was treated as an outlier and removed. Treatment replicate 4 from 48 hours and treatment replicate 1 from 96 hours were treated as outliers and removed. Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 28, 72, and 96 hours) in week 3.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	1	87.0	< 0.001
48hr	0	98.7	< 0.001
72hr	2	91.0	< 0.001
96hr	1	76.0	< 0.001

Week 4 results

For week 4, there were four replicates (1, 2, 3, 4) for both the control and treatment groups. Mortality was assessed at 24, 48, 72, and 96 hours.

Mortality of control replicates

During week 4, the following control replicates had mortality $\geq 10\%$:

Readings at 24 hours: Control replicate 2 had 52% mortality. This replicate was treated as an outlier.

Duration	Replicate	Dead	Total	%Mortality
48hr	2	3	25	12
96hr	4	4	25	16

Analysis of treatment replicates

During week 4, readings taken at 48 hours had significantly different mortality between replicates.

Readings at 48 hours: Replicate 2 (68% mortality) was significantly different from replicate 1 (100% mortality), replicate 3 (100% mortality), and replicate 4 (100% mortality). Replicate 2 was treated as an outlier.

Duration	Replicates for comparison	Adjusted p-value
48hr	1, 2	0.0242
48hr	2, 3	0.0242
48hr	2, 4	0.0242

Comparison of treatment to control

Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 48, 72, and 96 hours) in week 4.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	2	97	< 0.001
48hr	6	89	< 0.001
72hr	2	98	< 0.001

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
96hr	5	98	< 0.001

Comparison of treatment to control (excluding outliers)

Control replicate 2 from 24 hours was treated as an outlier and removed. Treatment replicate 2 from 48 hours was treated as an outlier and removed. Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 48, 72, and 96 hours) in week 4.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	2.0	97.0	< 0.001
48hr	4.0	85.3	< 0.001
72hr	2.0	98.0	< 0.001
96hr	1.3	98.0	< 0.001

Week 5 results

For week 5, there were four replicates (1, 2, 3, 4) for both the control and treatment groups. Mortality was assessed at 24 and 48 hours.

Mortality of control replicates

During week 5, the following control replicates had mortality $\geq 10\%$:

Readings at 24 hours: Control replicate 1 had 12% mortality. This replicate was treated as an outlier.

Readings at 48 hours: Control replicate 4 had 12% mortality. This replicate was treated as an outlier.

Duration	Replicate	Dead	Total	%Mortality
48hr	1	3	25	12

Analysis of treatment replicates

There were no treatment replicates for week 5 that were significantly different from other treatment replicates.

Comparison of treatment to control

Mortality was significantly greater in the treatment than in the control for each experiment duration (24 and 48 hours) in week 5.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	2	95	< 0.001
48hr	3	100	< 0.001

Comparison of treatment to control (excluding outliers)

Control replicate 1 from 24 hours and control replicate 4 from 48 hours were treated as outliers and removed. Mortality was significantly greater in the treatment than in the control for each experiment duration (24 and 48 hours) in week 5.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	2	95	< 0.001
48hr	0	100	< 0.001

Week 6 results

For week 6, there were four replicates (1, 2, 3, 4) for both the control and treatment groups. Mortality was assessed at 24, 48, and 72 hours.

Mortality of control replicates

During week 6, the following control replicates had mortality $\geq 10\%$:

Readings at 72 hours: Control replicate 3 had 16% mortality, and control replicate 4 had 12% mortality. These replicates were treated as outliers.

Duration	Replicate	Dead	Total	%Mortality
72hr	2	13	25	52
72hr	4	7	25	28

Analysis of treatment replicates

There were no treatment replicates for week 6 that were significantly different from other treatment replicates.

Comparison of treatment to control

Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 48, and 72 hours) in week 6.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	0	96	< 0.001
48hr	2	99	< 0.001
72hr	20	100	< 0.001

Comparison of treatment to control (excluding outliers)

Control replicates 3 and 4 from 72 hours were treated as outliers and removed. Mortality was significantly greater in the treatment than in the control for each experiment duration (24, 48, and 72 hours) in week 6.

Duration	Control (% Mortality)	Treatment (% Mortality)	p.value
24hr	0	96	< 0.001
48hr	2	99	< 0.001
72hr	0	100	< 0.001